

Bayesian Analysis for WSEI

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2     4.0.2      v tibble     3.3.1
v lubridate   1.9.5      v tidyr      1.3.2
v purrr       1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(lubridate)
library(brms)
```

```
Loading required package: Rcpp
Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').
```

```
Attaching package: 'brms'
```

```
The following object is masked from 'package:stats':
```

```
ar
```

```
library(bayesplot)
```

This is bayesplot version 1.15.0

- Online documentation and vignettes at mc-stan.org/bayesplot
- bayesplot theme set to bayesplot::theme_default()
 - * Does `_not_` affect other ggplot2 plots
 - * See `?bayesplot_theme_set` for details on theme setting

Attaching package: 'bayesplot'

The following object is masked from 'package:brms':

`rhat`

```
library(posterior)
```

This is posterior version 1.6.1

Attaching package: 'posterior'

The following object is masked from 'package:bayesplot':

`rhat`

The following objects are masked from 'package:stats':

`mad`, `sd`, `var`

The following objects are masked from 'package:base':

`%in%`, `match`

```
library(tidybayes)
```

Attaching package: 'tidybayes'

The following objects are masked from 'package:brms':

`dstudent_t`, `pstudent_t`, `qstudent_t`, `rstudent_t`

```
library(mgcv)
```

Loading required package: nlme

Attaching package: 'nlme'

The following object is masked from 'package:dplyr':

collapse

This is mgcv 1.9-4. For overview type '?mgcv'.

Attaching package: 'mgcv'

The following objects are masked from 'package:brms':

s, t2

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

1. Data

```
data_path <- "/Users/cy027/Desktop/merged_with_wsei.csv"
df_raw <- readr::read_csv(data_path, show_col_types = FALSE)

# Standardize names a bit (avoid spaces in formulas)
df <- df_raw %>%
  rename(
    station_id = `Station ID`,
    date = Date,
    lat = Latitude,
    lon = Longitude,
```

```

    aqi = AQI
  ) %>%
  mutate(
    date = as.Date(date),
    dow = wday(date, label = TRUE),
    month = month(date),
    doy = yday(date),
    year = year(date)
  )

# glimpse(df)
# summary(df$aqi)

```

Interpretation: this sets up the panel structure needed for hierarchical modelling (i.e. many days per station) and ensures the model can distinguish from day-to-day variation vs. station-to-station baseline differences.

1.1 Outcome choice

For **RQ1**, we model next-day AQI. That aligns with your “warning” goal and avoids same-day simultaneity.

```

df <- df %>%
  arrange(station_id, date) %>%
  group_by(station_id) %>%
  mutate(aqi_next = lead(aqi, 1)) %>%
  ungroup() %>%
  filter(!is.na(aqi_next))

```

Interpretation: this avoids same-day simultaneity (smoke today vs AQI today can be contemporaneous).

2. RQ1 - Bayesian hierarchical regression

Model specification

- Response: `aqi_next`
- Fixed effects: wildfire lags (k0–k3), meteorology, traffic, weekday
- Seasonality: smooth function of day-of-year (cyclic)

- Random intercept: station-level baseline differences

You can choose one wildfire intensity family as “primary” (recommended: **hfi**) and treat frp/tfc as sensitivity checks.

```
# Choose wildfire variables (primary)
wsei_vars <- c("wsei_hfi_k0", "wsei_hfi_k1", "wsei_hfi_k2", "wsei_hfi_k3")

# Basic covariates (edit based on what exists in your file)
covars <- c("MeanTemp", "TotalPrecip", "MaxTemp", "MinTemp", "TempRange", "TotalTraffic", "wind_dir_deg")

# Keep only columns we need and drop missing
df_m <- df %>%
  select(station_id, date, aqi_next, dow, doy, month, year, all_of(wsei_vars), all_of(covars),
  drop_na()

# Standardize continuous predictors for stable priors
scale_cols <- c(wsei_vars, covars)
df_m <- df_m %>%
  mutate(across(all_of(scale_cols), ~ as.numeric(scale(.x)), .names = "{.col}_z"))
```

cyclic seasonality under brms formula

```
f_rq1 <- bf(
  aqi_next ~
    wsei_hfi_k0_z + wsei_hfi_k1_z + wsei_hfi_k2_z + wsei_hfi_k3_z +
    MeanTemp_z + TotalPrecip_z + MaxTemp_z + MinTemp_z + TempRange_z +
    TotalTraffic_z + wind_dir_deg_z +
    dow +
    s(doy, bs = "cc") +
    (1 | station_id)
)

# Priors (work well when predictors are standardized)
priors <- c(
  prior(normal(0, 1), class = "b"),
  prior(normal(0, 3), class = "Intercept"),
  prior(exponential(1), class = "sd"),      # station random intercept sd
  prior(exponential(1), class = "sigma")    # residual sd
)
```

Fit with MCMC

```
fit_rq1 <- brm(
  formula = f_rq1,
  data = df_m,
  family = gaussian(),
  prior = priors,
  chains = 4, cores = 4, iter = 2000, warmup = 1000,
  seed = 946
)
```

Compiling Stan program...

Trying to compile a simple C file

```
Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
using C compiler: 'Apple clang version 17.0.0 (clang-1700.6.3.2)'
using SDK: 'MacOSX26.2.sdk'
clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -
DNDEBUG -I"/Users/cy027/Library/R/arm64/4.5/library/Rcpp/include/" -
I"/Users/cy027/Library/R/arm64/4.5/library/RcppEigen/include/" -I"/Users/cy027/Library/R/arm64/4.5/library/BH/include" -I"/Users/cy027/Library/R/arm64/4.5/library/StanHeaders/include/" -I"/Users/cy027/Library/R/arm64/4.5/library/rstan/include" -DEIGEN_NO_DEBUG -
DBOOST_DISABLE_ASSERTS -DBOOST_PENDING_INTEGER_LOG2_HPP -DSTAN_THREADS -
DUSE_STANC3 -DSTRICT_R_HEADERS -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION -
D_HAS_AUTO_PTR_ETC=0 -include '/Users/cy027/Library/R/arm64/4.5/library/StanHeaders/include/' -D_REENTRANT -DRCPP_PARALLEL_USE_TBB=1 -I/opt/R/arm64/include -fPIC -
falign-functions=64 -Wall -g -O2 -c foo.c -o foo.o
In file included from <built-in>:1:
In file included from /Users/cy027/Library/R/arm64/4.5/library/StanHeaders/include/stan/math/
In file included from /Users/cy027/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Dense
In file included from /Users/cy027/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Core:
/Users/cy027/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:
  679 | #include <cmath>
      |           ^~~~~~
1 error generated.
make: *** [foo.o] Error 1
```

Start sampling

```
fit_rq1
```

```
Family: gaussian
Links: mu = identity
Formula: aqi_next ~ wsei_hfi_k0_z + wsei_hfi_k1_z + wsei_hfi_k2_z + wsei_hfi_k3_z + MeanTemp
Data: df_m (Number of observations: 9885)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Smoothing Spline Hyperparameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sds(sday_1)	2.10	0.76	1.13	4.06	1.00	633	976

Multilevel Hyperparameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	20.55	2.75	15.71	26.21	1.01	638	1127

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	11.70	3.34	5.25	18.18	1.00	1683	1734
wsei_hfi_k0_z	1.20	0.23	0.74	1.64	1.00	3717	2883
wsei_hfi_k1_z	-0.67	0.31	-1.27	-0.04	1.00	3071	2613
wsei_hfi_k2_z	0.30	0.35	-0.36	1.00	1.00	2817	2446
wsei_hfi_k3_z	4.12	0.28	3.58	4.65	1.00	3495	3039
MeanTemp_z	0.61	0.82	-1.01	2.24	1.00	3812	3219
TotalPrecip_z	-1.77	0.15	-2.08	-1.47	1.00	4736	2878
MaxTemp_z	0.98	0.83	-0.68	2.58	1.00	3138	2787
MinTemp_z	0.27	0.79	-1.28	1.81	1.00	2724	2650
TempRange_z	1.94	0.32	1.32	2.57	1.00	2595	2564
TotalTraffic_z	-1.27	0.35	-1.97	-0.58	1.00	4579	2784
wind_dir_deg_z	-1.22	0.15	-1.51	-0.94	1.00	4856	2877
dow.L	-0.74	0.38	-1.47	-0.01	1.00	5151	3220
dow.Q	-1.84	0.36	-2.52	-1.14	1.00	5710	3318
dow.C	1.62	0.37	0.89	2.32	1.00	5064	2864
dowE4	-0.67	0.35	-1.36	0.05	1.00	5161	3151
dowE5	0.23	0.36	-0.47	0.94	1.00	5955	2953
dowE6	0.09	0.36	-0.64	0.77	1.00	4982	3065

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	14.56	0.11	14.35	14.78	1.00	6025	2716

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4:
Chain 4: Elapsed Time: 80.772 seconds (Warm-up)
Chain 4:           76.518 seconds (Sampling)
Chain 4:           157.29 seconds (Total)
Chain 4:
Chain 1: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1:
Chain 1: Elapsed Time: 77.464 seconds (Warm-up)
Chain 1:           84.196 seconds (Sampling)
Chain 1:           161.66 seconds (Total)
Chain 1:
Chain 3: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2:
Chain 2: Elapsed Time: 83.924 seconds (Warm-up)
Chain 2:           85.933 seconds (Sampling)
Chain 2:           169.857 seconds (Total)
Chain 2:
Chain 3: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3:
Chain 3: Elapsed Time: 80.912 seconds (Warm-up)
Chain 3:           101.076 seconds (Sampling)
Chain 3:           181.988 seconds (Total)
Chain 3:
```

Model

Gaussian hierarchical regression for next-day AQI with:

- Wildfire exposure lags: wsei_hfi_k0 ... wsei_hfi_k3
- Weather/traffic controls
- Weekday factor
- Cyclic seasonality spline `s(doy, bs="cc")`

- Station random intercept (1 | station_id)

Data size: 9885 observations across 19 stations

Key result: wildfire lags (the main story for RQ1)

From the regression coefficients table:

- **k0 (same-day exposure): +1.20 AQI** with 95% CrI [0.74, 1.64]

→ strong positive association.

- **k1: -0.67 AQI** with CrI [-1.27, -0.04]

→ small negative association (barely below zero)

- **k2: +0.30 AQI** with CrI [-0.36, 1.00]

→ uncertain (crosses zero)

- **k3: +4.12 AQI** with CrI [3.58, 4.65]

→ *very strong* and highly certain positive association.

How to explain it simply: After controlling for weather, seasonality, weekday, traffic, and station differences, the model finds wildfire exposure is associated with higher next-day AQI, and the strongest association appears around **3 days later**.

Controls: what's plausible vs what to be careful about

- **TotalPrecip: -1.77** (CrI [-2.08, -1.47]) → precipitation lowers AQI (physically plausible).
- **TempRange: +1.94** (CrI [1.32, 2.57]) → larger temp swings associated with higher AQI (plausible via boundary-layer/dispersion dynamics).
- **TotalTraffic: -1.27** (CrI [-1.97, -0.58]) → statistically strong but **not a causal claim**; could reflect confounding/measurement mismatch.

wind_dir_deg shows a strong negative coefficient, but wind direction is circular; using degrees linearly can be misleading (good “future refinement” item).

Hierarchical part (station effects)

- **sd(Intercept) 20.55** (CrI [15.71, 26.21]) → stations have very different baseline AQI levels; hierarchical structure is warranted.

- **sigma** 14.56 → remaining unexplained day-to-day variation in AQI_{next}.

Fit **Student's t hierarchical regression** for next-day AQI with

```
# Student-t likelihood (robust to spikes/heavy tails)
f_rq1_t <- bf(
  aqi_next ~
    wsei_hfi_k0_z + wsei_hfi_k1_z + wsei_hfi_k2_z + wsei_hfi_k3_z +
    MeanTemp_z + TotalPrecip_z + MaxTemp_z + MinTemp_z + TempRange_z +
    TotalTraffic_z + wind_dir_deg_z +
    dow +
    s(doy, bs = "cc") +
    (1 | station_id)
)

priors_t <- c(
  prior(normal(0, 1), class = "b"),
  prior(normal(0, 5), class = "Intercept"),
  prior(exponential(1), class = "sd"),
  prior(exponential(1), class = "sigma"),
  # df (nu) controls tail-heaviness; this prior encourages heavier tails than Gaussian when
  prior(gamma(2, 0.1), class = "nu")
)

fit_rq1_student <- brm(
  formula = f_rq1_t,
  data = df_m,
  family = student(),      # <-- key change vs gaussian()
  prior = priors_t,
  chains = 4, cores = 4,
  iter = 3000, warmup = 1500,
  control = list(adapt_delta = 0.95),
  seed = 946
)
```

Compiling Stan program...

Trying to compile a simple C file

Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
 using C compiler: 'Apple clang version 17.0.0 (clang-1700.6.3.2)'
 using SDK: 'MacOSX26.2.sdk'

```

clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -
DNDEBUG -I"/Users/cy027/Library/R/arm64/4.5/library/Rcpp/include/" -
I"/Users/cy027/Library/R/arm64/4.5/library/RcppEigen/include/" -I"/Users/cy027/Library/R/arm64/4.5/library/BH/include" -I"/Users/cy027/Library/R/arm64/4.5/library/StanHeaders/include/" -I"/Users/cy027/Library/R/arm64/4.5/library/rstan/include" -DEIGEN_NO_DEBUG -
DBOOST_DISABLE_ASSERTS -DBOOST_PENDING_INTEGER_LOG2_HPP -DSTAN_THREADS -
DUSE_STANC3 -DSTRICT_R_HEADERS -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION -
D_HAS_AUTO_PTR_ETC=0 -include '/Users/cy027/Library/R/arm64/4.5/library/StanHeaders/include/' -D_REENTRANT -DRCPP_PARALLEL_USE_TBB=1 -I/opt/R/arm64/include -fPIC -
falign-functions=64 -Wall -g -O2 -c foo.c -o foo.o
In file included from <built-in>:1:
In file included from /Users/cy027/Library/R/arm64/4.5/library/StanHeaders/include/stan/math:
In file included from /Users/cy027/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Dense:
In file included from /Users/cy027/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Core:
/Users/cy027/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:
  679 | #include <cmath>
      |           ^~~~~~
1 error generated.
make: *** [foo.o] Error 1

```

Start sampling

```
summary(fit_rq1_student)
```

```

Family: student
Links: mu = identity
Formula: aqi_next ~ wsei_hfi_k0_z + wsei_hfi_k1_z + wsei_hfi_k2_z + wsei_hfi_k3_z + MeanTemp
Data: df_m (Number of observations: 9885)
Draws: 4 chains, each with iter = 3000; warmup = 1500; thin = 1;
       total post-warmup draws = 6000

```

Smoothing Spline Hyperparameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sds(sdoy_1)	1.57	0.53	0.87	2.86	1.00	1270	2160

Multilevel Hyperparameters:

```

~station_id (Number of levels: 19)
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)    3.65      0.67    2.61    5.16 1.00    1379    2484

```

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	37.77	0.97	35.65	39.48	1.00	903	1596
wsei_hfi_k0_z	0.41	0.33	-0.21	1.08	1.00	5332	4688
wsei_hfi_k1_z	1.39	0.47	0.48	2.33	1.00	4393	4433
wsei_hfi_k2_z	-1.49	0.89	-3.20	0.14	1.00	3594	4321
wsei_hfi_k3_z	5.25	0.50	4.26	6.22	1.00	5884	4719
MeanTemp_z	0.64	0.82	-0.98	2.26	1.00	5597	4048
TotalPrecip_z	-1.66	0.12	-1.91	-1.42	1.00	8382	4630
MaxTemp_z	1.03	0.82	-0.57	2.64	1.00	4882	4461
MinTemp_z	0.37	0.77	-1.17	1.88	1.00	4035	4210
TempRange_z	1.84	0.30	1.25	2.43	1.00	4206	4131
TotalTraffic_z	-0.66	0.31	-1.27	-0.08	1.00	4906	4315
wind_dir_deg_z	-1.10	0.12	-1.34	-0.86	1.00	7462	4027
dow.L	-0.54	0.30	-1.12	0.05	1.00	7276	4354
dow.Q	-0.84	0.29	-1.41	-0.26	1.00	8878	4559
dow.C	1.01	0.30	0.43	1.60	1.00	8709	4545
dowE4	-0.63	0.30	-1.23	-0.03	1.00	7919	4352
dowE5	0.49	0.29	-0.10	1.07	1.00	8444	4559
dowE6	-0.38	0.30	-0.96	0.19	1.00	8310	4603

Further Distributional Parameters:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	9.72	0.12	9.48	9.95	1.00	5928	4385
nu	3.92	0.16	3.62	4.24	1.00	6030	4468

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
: 2700 / 3000 [ 90%] (Sampling)
Chain 1: Iteration: 2400 / 3000 [ 80%] (Sampling)
Chain 4: Iteration: 3000 / 3000 [100%] (Sampling)
Chain 4:
Chain 4: Elapsed Time: 114.682 seconds (Warm-up)
Chain 4: 82.696 seconds (Sampling)
Chain 4: 197.378 seconds (Total)
Chain 4:
Chain 2: Iteration: 2700 / 3000 [ 90%] (Sampling)
Chain 3: Iteration: 3000 / 3000 [100%] (Sampling)
Chain 3:
Chain 3: Elapsed Time: 110.627 seconds (Warm-up)
Chain 3: 98.171 seconds (Sampling)
Chain 3: 208.798 seconds (Total)
Chain 3:
```

```

Chain 1: Iteration: 2700 / 3000 [ 90%] (Sampling)
Chain 2: Iteration: 3000 / 3000 [100%] (Sampling)
Chain 2:
Chain 2: Elapsed Time: 120.976 seconds (Warm-up)
Chain 2:           104.204 seconds (Sampling)
Chain 2:           225.18 seconds (Total)
Chain 2:
Chain 1: Iteration: 3000 / 3000 [100%] (Sampling)
Chain 1:
Chain 1: Elapsed Time: 119.27 seconds (Warm-up)
Chain 1:           111.093 seconds (Sampling)
Chain 1:           230.363 seconds (Total)
Chain 1:

```

Student-t model interpretation (RQ1: next-day AQI)

Model and sampling quality

- **Family:** `student` with identity link for the mean.
- **N = 9,885, 19 stations**, 4 chains, 6,000 post-warmup draws.
- **Rhat = 1.00** across parameters, ESS is large → sampling/convergence look healthy.

This is a statistically credible fit.

What Student-t is doing here (key insight)

The Student-t likelihood has an extra parameter **nu** controlling tail-heaviness:

- **nu = 3.92** (95% CrI: 3.62–4.24)

That’s *very heavy-tailed* (a Normal is like $\text{nu} \rightarrow \infty$).

Interpretation: your data has outliers/spikes, and the model agrees—extremes are common enough that a Gaussian error model is too “thin-tailed”.

Also:

- **sigma = 9.72** (CrI: 9.48–9.95)

This sigma is the scale parameter under Student-t; it’s expected to be **smaller than Gaussian sigma** because heavy tails “explain” extremes via nu rather than by inflating sigma.

Station heterogeneity

- $\text{sd}(\text{Intercept}) = 3.65$ (CrI: 2.61–5.16)

This is the between-station baseline variability in AQI_{next} after controls. (It's smaller than your Gaussian model's station SD because the model is now attributing less variance to persistent station offsets and more to the heavy-tailed residual component; also depends on which sample survived NA drops in each run.)

Seasonality strength

- $\text{sds}(\text{sday}_1) = 1.57$ (CrI: 0.87–2.86)
- The day-of-year cyclic smooth contributes meaningfully; seasonality is real and non-negligible.

2) Student-t coefficients: wildfire and controls

Important: predictors were z-scored, outcome is in AQI units.

So each coefficient is **AQI units change in next-day AQI per +1 SD increase** in that predictor.

Wildfire lag effects (WSEI HFI)

- **k3: 5.25** (CrI: 4.26–6.22) strong positive
- **k1: 1.39** (CrI: 0.48–2.33) positive
- **k0: 0.41** (CrI: –0.21–1.08) crosses 0 (weak evidence)
- **k2: –1.49** (CrI: –3.20–0.14) mostly negative but crosses 0 slightly

Interpretation: wildfire exposure is strongly associated with higher next-day AQI, with the dominant signal at **lag 3**, and some evidence at **lag 1**. This is consistent with a delayed transport/persistence story.

The mixed lag signs (k2 negative-ish) are common in distributed lag models because lagged WSEIs are correlated and coefficients can “trade off”. It does **not** mean wildfire reduces AQI on lag 2 in a causal sense.

Controls

- **TotalPrecip:** **−1.66** (CrI: −1.91 to −1.42) strong negative

Physically plausible: precipitation scavenges particles / improves air quality.

- **TempRange:** **+1.84** (CrI: 1.25–2.43) positive

Suggests dispersion/boundary-layer conditions matter.

- **wind_dir_deg:** **−1.10** (CrI: −1.34 to −0.86) statistically strong, but interpret carefully

Wind direction is circular; linear degrees can produce misleading signs. Better to use sin/cos in your next revision.

- **TotalTraffic:** **−0.66** (CrI: −1.27 to −0.08) statistically negative
- Again: don't interpret causally; likely confounding/measurement mismatch.
- MeanTemp/MaxTemp/MinTemp are uncertain (intervals cross 0), which is fine.

3) Gaussian vs Student-t: what changed, what stayed the same

A) Tail behavior and uncertainty: Student-t wins (for realism)

- Gaussian model's residual SD was ~**14.56** (from your earlier output).
- Student-t scale **sigma = 9.72**, **nu = 3.9** (**heavy tails**).

Meaning: Gaussian is forced to inflate σ to account for extremes.

Student-t explains extremes with heavy tails (nu small), letting σ reflect typical variation.

This typically improves:

- posterior predictive checks in the tails,
- robustness of coefficients to extreme days,
- interpretability for “normal days vs extreme days”.

B) Wildfire signal: consistent direction, stronger/clearer in Student-t

Both models tell the same story: **lag-3 wildfire exposure is strongly positive**.

- Student-t k3: **+5.25 AQI** (CrI 4.26–6.22)
- Earlier Gaussian k3 was also strongly positive (you previously had $\sim +4.12$ AQI with a tight CrI in that run).

Student-t is giving a *somewhat larger* k3 magnitude here. That can happen because:

- outliers have less leverage, which can change coefficient allocation,
- different likelihood changes how the model attributes variance,
- slightly different rows may remain after `drop_na()`.

The key is: **sign and significance are stable**, which is what you want to show robustness.

C) Model “fit quality” comparison (what you should check next)

To make the comparison rigorous in your report/presentation, compute:

- **LOO / WAIC** (expected out-of-sample predictive accuracy)
- posterior predictive checks overlay for tails
- In many real AQI datasets, Student-t often wins on LOO when extremes exist.

4) What to say in your boss presentation

Slide: “Why we moved from Gaussian to Student-t”

- AQI has rare but large spikes (smoke episodes). A Gaussian error model assumes thin tails and can misrepresent uncertainty.
- The Student-t fit estimates **ν 3.9**, indicating heavy tails are real in our data, and improves robustness.”

Slide: “Core wildfire finding is robust”

- Across model variants, wildfire exposure shows a strong delayed association with next-day AQI.
- The dominant effect is around **lag 3**, consistent with smoke transport/persistence.

Slide: “Operational implication”

- This supports using a 0–3 day lagged wildfire risk signal, with particular attention to the 3-day lag.
- We can use the risk model (RQ3) to translate this into warning probability.

Slide: “What we still won’t claim”

- These are attribution-style associations, not causal effects.
- Wind direction needs circular encoding; lag coefficients can trade off due to correlation.

3. Assumption Check

3.1 Convergence

```
summary(fit_rq1)$fixed %>% as.data.frame()
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	11.69847034	3.3419072	5.2480142	18.179288057	1.0009635	1682.865
wsei_hfi_k0_z	1.19996411	0.2286449	0.7432221	1.644013144	1.0004838	3717.387
wsei_hfi_k1_z	-0.67219947	0.3132830	-1.2707414	-0.037085634	1.0001140	3070.856
wsei_hfi_k2_z	0.30285156	0.3466422	-0.3622679	0.995811864	1.0015246	2816.582
wsei_hfi_k3_z	4.11737893	0.2762992	3.5797039	4.653087774	1.0004462	3495.419
MeanTemp_z	0.60699930	0.8202809	-1.0083709	2.239472877	1.0001828	3812.102
TotalPrecip_z	-1.77432899	0.1536979	-2.0762484	-1.473038873	1.0003927	4736.172
MaxTemp_z	0.98441105	0.8335719	-0.6759387	2.577359427	1.0000690	3138.293
MinTemp_z	0.27197969	0.7903411	-1.2817511	1.805004569	1.0007853	2723.803
TempRange_z	1.93867025	0.3226165	1.3227963	2.565563464	1.0011768	2595.189
TotalTraffic_z	-1.27201099	0.3520890	-1.9669730	-0.584174666	1.0010540	4579.034
wind_dir_deg_z	-1.22092808	0.1474706	-1.5110363	-0.938134314	0.9999617	4855.709
dow.L	-0.74212842	0.3757387	-1.4660650	-0.009432542	1.0000952	5151.392
dow.Q	-1.84392670	0.3564099	-2.5238642	-1.141095322	1.0009367	5710.229
dow.C	1.62464710	0.3681250	0.8914496	2.324951925	1.0010049	5064.103
dowE4	-0.67260544	0.3534306	-1.3551569	0.051934149	1.0002329	5161.020
dowE5	0.22840609	0.3581086	-0.4699597	0.935444255	1.0022489	5955.069
dowE6	0.08860044	0.3557793	-0.6416485	0.774077057	1.0011698	4981.931
Tail_ESS						
Intercept	1733.746					
wsei_hfi_k0_z	2882.615					
wsei_hfi_k1_z	2613.197					

```

wsei_hfi_k2_z  2445.822
wsei_hfi_k3_z  3038.753
MeanTemp_z     3219.479
TotalPrecip_z  2877.755
MaxTemp_z      2787.344
MinTemp_z      2650.152
TempRange_z    2564.387
TotalTraffic_z 2784.121
wind_dir_deg_z 2876.855
dow.L          3220.163
dow.Q          3318.349
dow.C          2864.298
dowE4          3151.054
dowE5          2953.275
dowE6          3064.782

```

```

# R-hat / ESS overview
posterior::summarise_draws(as_draws_df(fit_rq1)) %>%
  select(variable, rhat, ess_bulk, ess_tail) %>%
  filter(str_detect(variable, "^b_")) %>%
  arrange(desc(rhat)) %>%
  head(15)

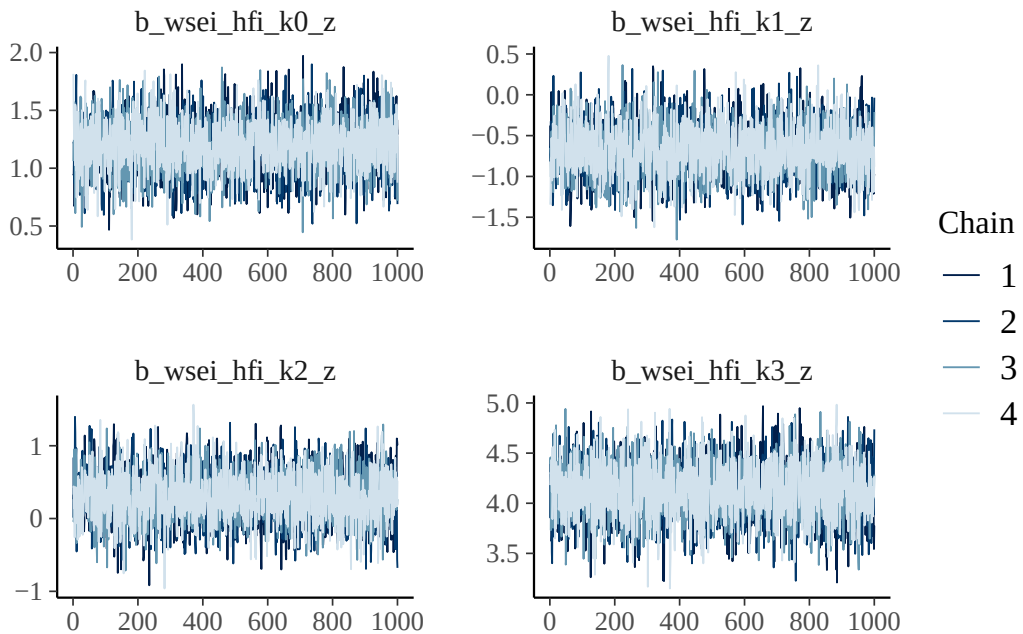
```

```

# A tibble: 15 x 4
  variable      rhat ess_bulk ess_tail
  <chr>        <dbl>   <dbl>   <dbl>
1 b_dowE5      1.00    5955.    2953.
2 b_wsei_hfi_k2_z 1.00    2817.    2446.
3 b_TempRange_z 1.00    2595.    2564.
4 b_dowE6      1.00    4982.    3065.
5 b_TotalTraffic_z 1.00    4579.    2784.
6 b_dow.C      1.00    5064.    2864.
7 b_Intercept  1.00    1683.    1734.
8 b_dow.Q      1.00    5710.    3318.
9 b_MinTemp_z  1.00    2724.    2650.
10 b_wsei_hfi_k0_z 1.00    3717.    2883.
11 b_wsei_hfi_k3_z 1.00    3495.    3039.
12 b_TotalPrecip_z 1.00    4736.    2878.
13 b_dowE4      1.00    5161.    3151.
14 b_MeanTemp_z  1.00    3812.    3219.
15 b_wsei_hfi_k1_z 1.00    3071.    2613.

```

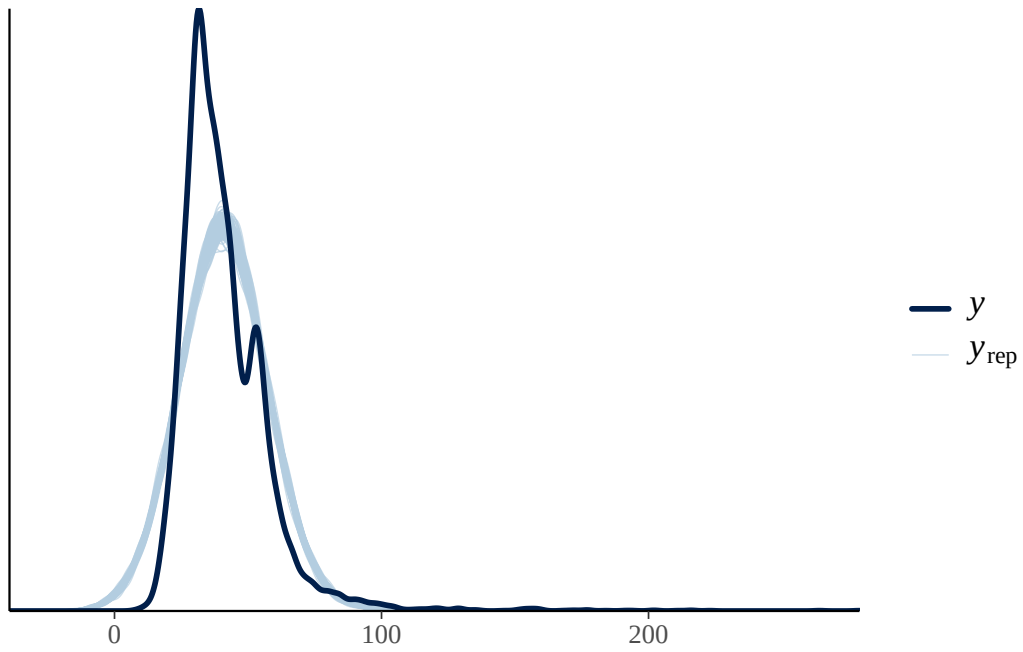
```
# Trace the plot
mcmc_trace(as_draws_array(fit_rq1), pars = c(
  "b_wsei_hfi_k0_z", "b_wsei_hfi_k1_z", "b_wsei_hfi_k2_z", "b_wsei_hfi_k3_z"
))
```



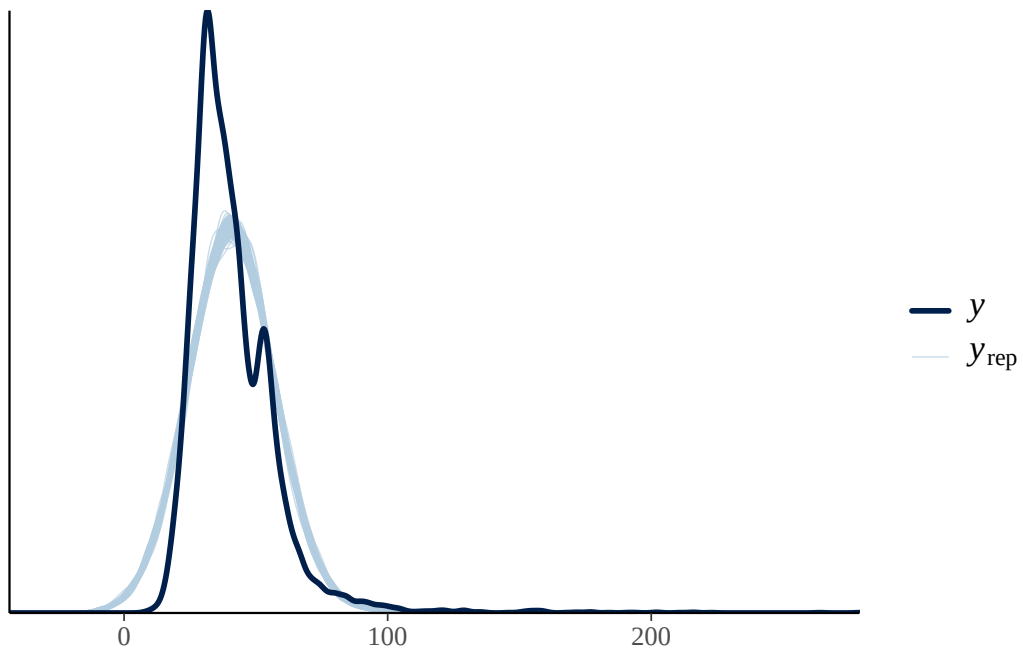
R-hat and ESS; they are essentially ~ 1.00 with large ESS for key coefficients. Trace plot for wildfire coefficients looks stable (chains mix well). Analysis **Interpretation:** the MCMC sampling is reliable enough to trust the posterior summaries.

3.2 Posterior Predictive Check

```
pp_check(fit_rq1, ndraws = 200)
```



```
pp_check(fit_rq1, type = "dens_overlay", ndraws = 100)
```



The overlay plots show the model captures the bulk distribution, but tails aren't perfect (you can see mismatch in the far right tail in the overlay).

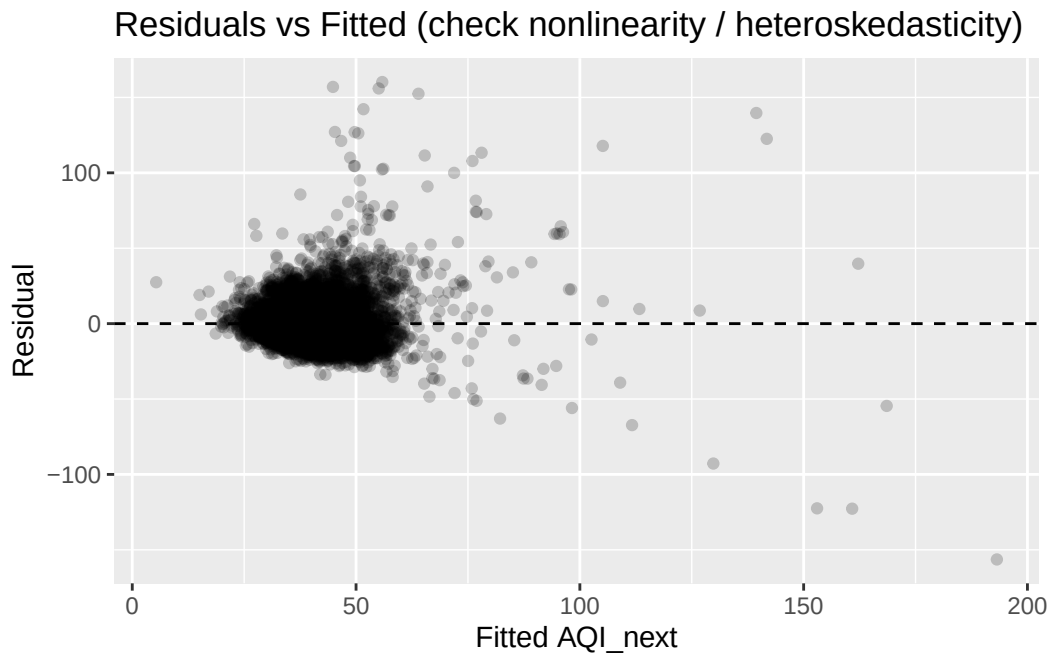
Interpretation: Gaussian errors are “okay for mean behavior,” but if extreme AQI accuracy matters, consider Student-t likelihood or tail modeling later.

3.3 Nonlinearity Check

```
res <- residuals(fit_rq1)
fit <- fitted(fit_rq1[, "Estimate"])

diag_df <- df_m %>%
  mutate(resid = res[, "Estimate"], fitted = fit)

ggplot(diag_df, aes(fitted, resid)) +
  geom_point(alpha = 0.2) +
  geom_hline(yintercept = 0, linetype = 2) +
  labs(title = "Residuals vs Fitted (check nonlinearity / heteroskedasticity)",
       x = "Fitted AQI_next", y = "Residual")
```



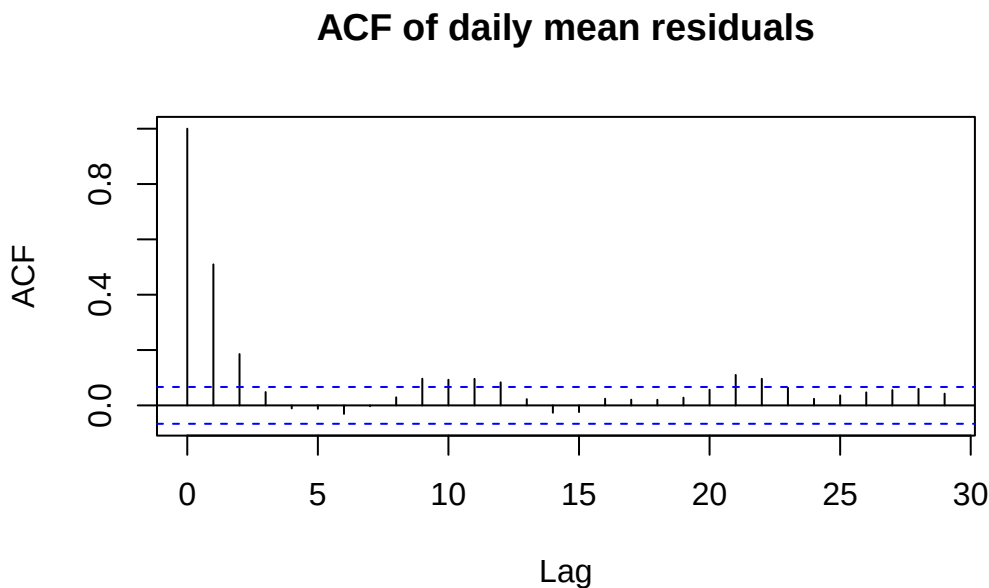
Residuals vs fitted plot shows variance widening at higher fitted AQI (fanning).

Interpretation: errors are larger on bad-air days; uncertainty is not constant.

Our model is directionally right, but uncertainty grows in high-AQI regimes—consistent with CS1’s ‘extreme days are harder’ limitation.

3.4 Autocorrelation

```
daily_res <- diag_df %>%  
  group_by(date) %>%  
  summarise(mean_resid = mean(resid), .groups = "drop") %>%  
  arrange(date)  
  
acf(daily_res$mean_resid, main = "ACF of daily mean residuals")
```



The ACF of daily mean residuals shows residual structure at short lags. **Interpretation:** there’s still time dependence not fully captured. If forecasting is the priority, adding lagged AQI or AR structure could improve predictive performance.

4. Wildfire Lag Effects

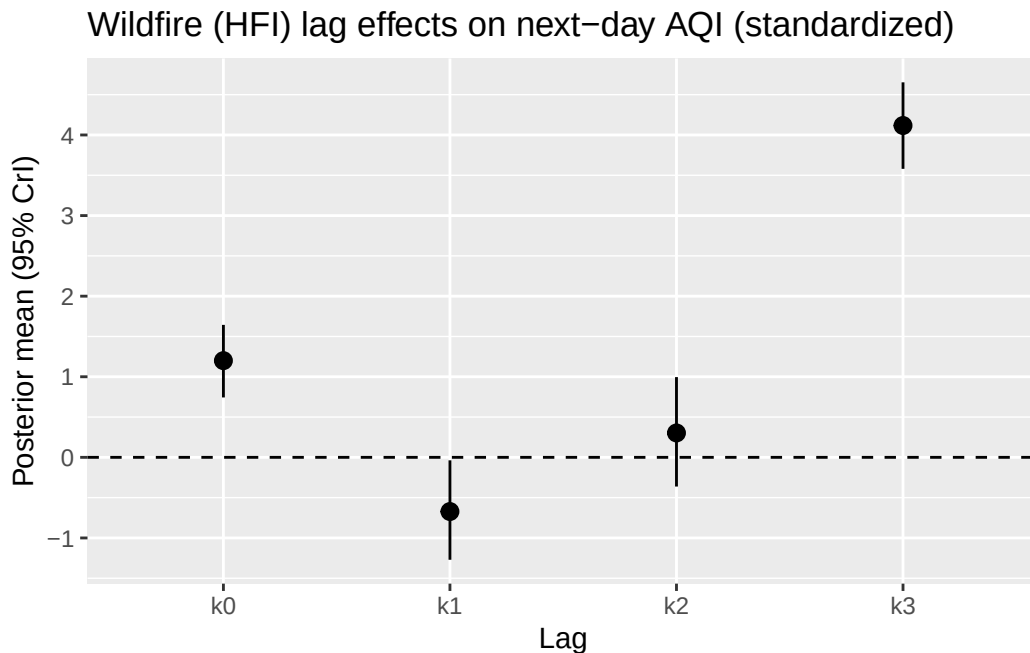
4.1 Posterior intervals for lag coefficients

```
coef_df <- as_draws_df(fit_rq1) %>%
  select(starts_with("b_wsei_hfi_k")) %>%
  pivot_longer(everything(), names_to = "term", values_to = "draw") %>%
  mutate(lag = str_extract(term, "k[0-3]"))
```

Warning: Dropping 'draws_df' class as required metadata was removed.

```
sum_df <- coef_df %>%
  group_by(lag) %>%
  summarise(
    mean = mean(draw),
    lo = quantile(draw, 0.025),
    hi = quantile(draw, 0.975),
    p_pos = mean(draw > 0),
    .groups = "drop"
  )

ggplot(sum_df, aes(x = lag, y = mean, ymin = lo, ymax = hi)) +
  geom_pointrange() +
  geom_hline(yintercept = 0, linetype = 2) +
  labs(title = "Wildfire (HFI) lag effects on next-day AQI (standardized)",
       y = "Posterior mean (95% CrI)", x = "Lag")
```



This plot is the “money plot” for RQ1 — it visualizes the lag coefficients with 95% credible intervals. It confirms:

- strong positive at k0
- uncertain at k2
- dominant positive at k3

4.2 Cumulative short-lag effect (0–3 days)

```
draws <- as_draws_df(fit_rq1)
cum <- with(draws,
             b_wsei_hfi_k0_z + b_wsei_hfi_k1_z + b_wsei_hfi_k2_z + b_wsei_hfi_k3_z)

tibble(cum = cum) %>%
  summarise(
    mean = mean(cum),
    lo = quantile(cum, 0.025),
    hi = quantile(cum, 0.975),
    p_pos = mean(cum > 0)
  )
```

```
# A tibble: 1 x 4
  mean    lo    hi p_pos
<dbl> <dbl> <dbl> <dbl>
1  4.95  4.63  5.27     1
```

You computed the sum of k0–k3 posterior draws **mean 4.95**, CrI [4.63, 5.27], and **p_pos = 1**

Interpretation: across the 0–3 day window, the total wildfire association with next-day AQI is almost certainly positive and large. This is very clean to communicate.

5. Operational Wildfire Signal

We tried to forecast the probability of high-AQI next day.

According to Air Quality Ontario, **Special Air Quality Statement (SAQS)** will be issued when the AQHI is forecast to reach or exceed **7 (High Risk)**.


```
# High AQI next day = 1 if next-day AQI is in the top 10% of days in the dataset
q90 <- quantile(df_m$aqi_next, probs = 0.90, na.rm = TRUE)
q95 <- quantile(df_m$aqi_next, probs = 0.95, na.rm = TRUE) # optional sensitivity

df_m <- df_m %>%
  mutate(
    high_next_90 = as.integer(aqi_next >= q90),
    high_next_95 = as.integer(aqi_next >= q95)
  )
```

Fit a Bayesian hierarchical logistic model

```
f_rq3_90 <- bf(
  high_next_90 ~
    wsei_hfi_k0_z + wsei_hfi_k1_z + wsei_hfi_k2_z + wsei_hfi_k3_z +
    MeanTemp_z + TotalPrecip_z + MaxTemp_z + MinTemp_z + TempRange_z +
    TotalTraffic_z + wind_dir_deg_z +
    dow +
    s(doy, bs = "cc") +
    (1 | station_id)
)

fit_rq3_90 <- brm(
  formula = f_rq3_90,
  data = df_m,
  family = bernoulli(link = "logit"),
  prior = c(
    prior(normal(0, 1), class = "b"),
    prior(normal(0, 2), class = "Intercept"),
    prior(exponential(1), class = "sd")
  ),
  chains = 4, cores = 4, iter = 2000, warmup = 1000,
  control = list(adapt_delta = 0.95), # helps avoid divergences
  seed = 946
)
```

Compiling Stan program...

Trying to compile a simple C file

Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c

```

using C compiler: 'Apple clang version 17.0.0 (clang-1700.6.3.2)'
using SDK: 'MacOSX26.2.sdk'
clang -arch arm64 -std=gnu2x -I"/Library/Frameworks/R.framework/Resources/include" -
DNDEBUG -I"/Users/cy027/Library/R/arm64/4.5/library/Rcpp/include/" -
I"/Users/cy027/Library/R/arm64/4.5/library/RcppEigen/include/" -I"/Users/cy027/Library/R/arm64/4.5/library/BH/include" -I"/Users/cy027/Library/R/arm64/4.5/library/StanHeaders/include/" -I"/Users/cy027/Library/R/arm64/4.5/library/rstan/include" -DEIGEN_NO_DEBUG -
DBOOST_DISABLE_ASSERTS -DBOOST_PENDING_INTEGER_LOG2_HPP -DSTAN_THREADS -
DUSE_STANC3 -DSTRICT_R_HEADERS -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION -
D_HAS_AUTO_PTR_ETC=0 -include '/Users/cy027/Library/R/arm64/4.5/library/StanHeaders/include' -D_REENTRANT -DRCPP_PARALLEL_USE_TBB=1 -I/opt/R/arm64/include -fPIC -
falign-functions=64 -Wall -g -O2 -c foo.c -o foo.o
In file included from <built-in>:1:
In file included from /Users/cy027/Library/R/arm64/4.5/library/StanHeaders/include/stan/math:
In file included from /Users/cy027/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Dense:
In file included from /Users/cy027/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/Core:
/Users/cy027/Library/R/arm64/4.5/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:
  679 | #include <cmath>
      |           ^~~~~~
1 error generated.
make: *** [foo.o] Error 1

```

Start sampling

fit_rq3_90

```

Family: bernoulli
Links: mu = logit
Formula: high_next_90 ~ wsei_hfi_k0_z + wsei_hfi_k1_z + wsei_hfi_k2_z + wsei_hfi_k3_z + Mean
Data: df_m (Number of observations: 9885)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000

```

Smoothing Spline Hyperparameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sds(sday_1)	0.34	0.11	0.20	0.63	1.01	789	1694

Multilevel Hyperparameters:

```

~station_id (Number of levels: 19)

```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.63	0.13	0.43	0.93	1.00	1100	2017

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-2.55	0.15	-2.86	-2.25	1.00	879	1600
wsei_hfi_k0_z	-0.07	0.04	-0.16	0.01	1.00	3291	2997
wsei_hfi_k1_z	0.17	0.06	0.05	0.29	1.00	2653	2630
wsei_hfi_k2_z	-0.17	0.07	-0.31	-0.03	1.00	2522	2574
wsei_hfi_k3_z	0.45	0.07	0.32	0.59	1.00	3234	2895
MeanTemp_z	0.19	0.82	-1.45	1.75	1.00	3065	2535
TotalPrecip_z	-0.43	0.06	-0.55	-0.33	1.00	4545	3039
MaxTemp_z	0.34	0.84	-1.36	1.95	1.00	2661	2558
MinTemp_z	0.18	0.77	-1.34	1.68	1.00	2204	2462
TempRange_z	0.41	0.28	-0.15	0.97	1.00	2173	2105
TotalTraffic_z	-0.14	0.07	-0.29	0.01	1.00	2337	2352
wind_dir_deg_z	-0.26	0.04	-0.33	-0.19	1.00	4419	2527
dow.L	-0.23	0.10	-0.42	-0.03	1.00	5232	3130
dow.Q	-0.45	0.10	-0.65	-0.25	1.00	4280	2896
dow.C	0.23	0.09	0.05	0.41	1.00	3938	3075
dowE4	-0.21	0.09	-0.39	-0.04	1.00	4663	2754
dowE5	0.15	0.09	-0.02	0.33	1.00	4496	3157
dowE6	-0.17	0.09	-0.34	0.00	1.00	4827	2884

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

(Sampling)

```
Chain 3: Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1: Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4: Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2: Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3:
Chain 3: Elapsed Time: 118.643 seconds (Warm-up)
Chain 3: 118.307 seconds (Sampling)
Chain 3: 236.95 seconds (Total)
```

```

Chain 3:
Chain 1: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1:
Chain 1: Elapsed Time: 127.657 seconds (Warm-up)
Chain 1:           112.8 seconds (Sampling)
Chain 1:           240.457 seconds (Total)
Chain 1:
Chain 4: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4:
Chain 4: Elapsed Time: 123.278 seconds (Warm-up)
Chain 4:           126.731 seconds (Sampling)
Chain 4:           250.009 seconds (Total)
Chain 4:
Chain 2: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2:
Chain 2: Elapsed Time: 120.288 seconds (Warm-up)
Chain 2:           132.527 seconds (Sampling)
Chain 2:           252.815 seconds (Total)
Chain 2:

```

Logistic hierarchical model - fit Bernoulli likelihood with logit link

- same covariates + station random intercept + seasonality spline

These are **log-odds** effects (per 1 SD increase), not AQI units.

- **k3: 0.45** CrI [0.32, 0.59] → strong positive predictor of extreme-day risk Analysis

Odds ratio $\exp(0.45)=1.57 \rightarrow \sim 57\%$ higher odds per +1 SD.

- **k1: 0.17** CrI [0.05, 0.29] → smaller positive effect
- **k2: -0.17** CrI [-0.31, -0.03] → small negative (possible lag tradeoff / multicollinearity)
- **k0 crosses 0** (not strong evidence)

How to explain to stakeholders: The model's wildfire exposure features, especially the 3-day lag, meaningfully increase the probability of a high-AQI day tomorrow.

5.1 Risk score and calibration

Compute posterior mean risk and do a simple calibration curve by deciles.

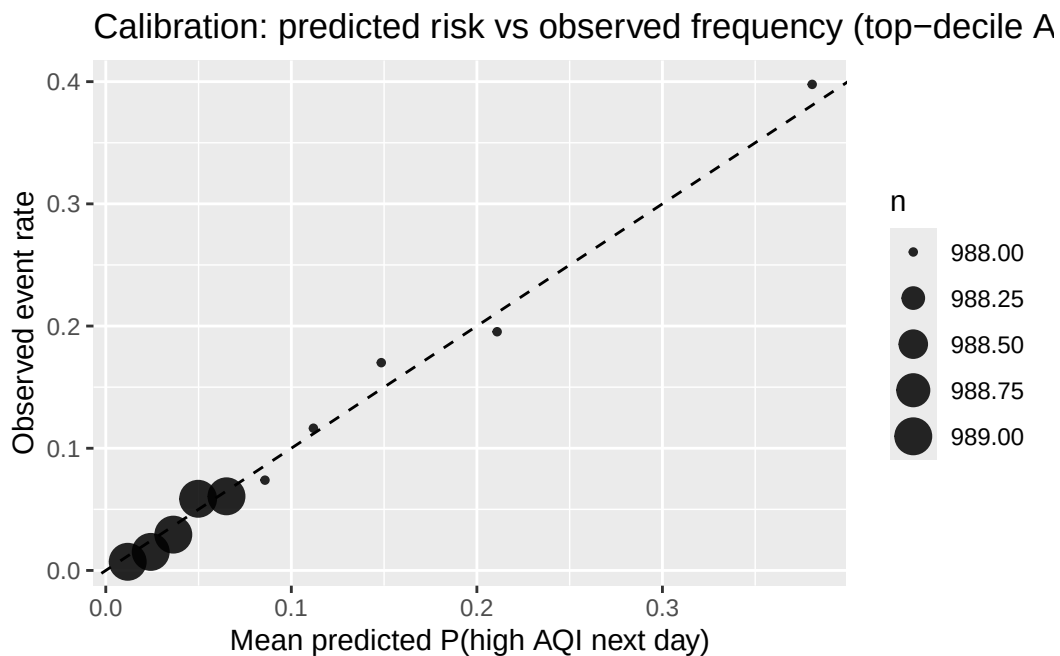
```

# posterior expected probability
p_hat <- fitted(fit_rq3_90, scale = "response")[, "Estimate"]
df_risk <- df_m %>% mutate(p_risk = p_hat)

cal <- df_risk %>%
  mutate(bin = ntile(p_risk, 10)) %>%
  group_by(bin) %>%
  summarise(
    mean_pred = mean(p_risk),
    obs_rate = mean(high_next_90),
    n = n(),
    .groups = "drop"
  )

ggplot(cal, aes(mean_pred, obs_rate, size = n)) +
  geom_point(alpha = 0.85) +
  geom_abline(slope = 1, intercept = 0, linetype = 2) +
  labs(
    title = "Calibration: predicted risk vs observed frequency (top-decile AQI event)",
    x = "Mean predicted P(high AQI next day)",
    y = "Observed event rate"
  )

```



You bin predictions into deciles and compare: mean predicted probability vs observed event rate

The points lie fairly close to the diagonal in your plot, which supports that the risk score is not wildly miscalibrated.

How to present: When the model says 20% risk, historically about ~20% of those days are truly high-AQI.

5.2 “Wildfire-driven risk” diagnostic

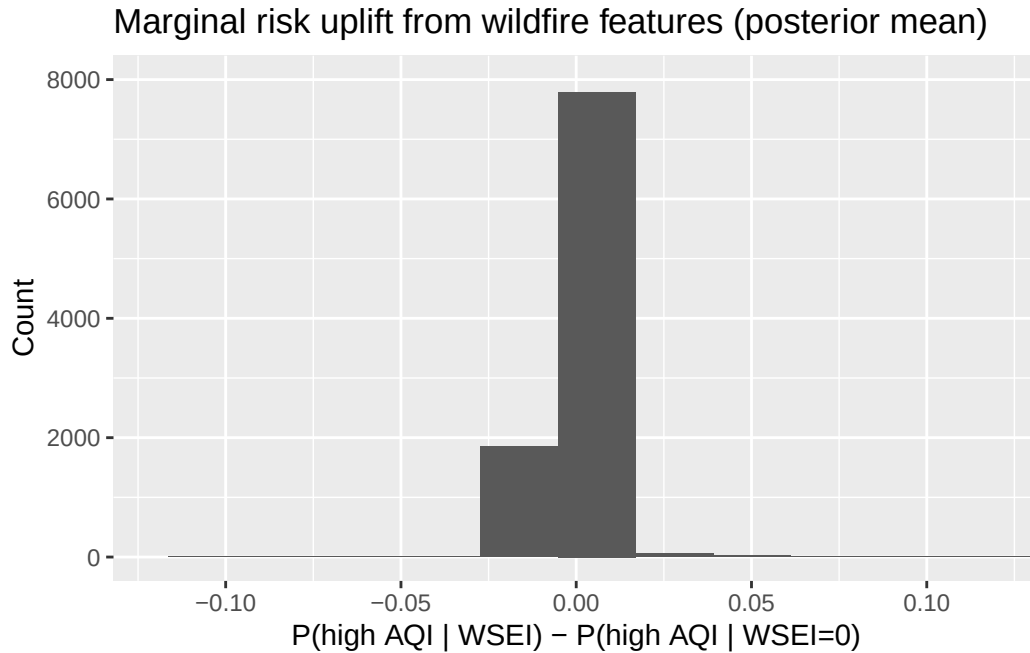
Operationally useful check: compare risk with and without wildfire inputs.

```
# Create two prediction datasets: with true WSEI vs WSEI set to 0
new_with <- df_m
new_wo    <- df_m %>% mutate(across(contains("wsei_hfi_k"), ~ 0))

p_with <- fitted(fit_rq3_90, newdata = new_with, scale = "response")[, "Estimate"]
p_wo   <- fitted(fit_rq3_90, newdata = new_wo, scale = "response")[, "Estimate"]

delta <- tibble(delta_risk = p_with - p_wo)

ggplot(delta, aes(delta_risk)) +
  geom_histogram(bins = 50) +
  coord_cartesian(xlim = c(-0.12, 0.12), ylim = c(0, 8000)) +
  labs(
    title = "Marginal risk uplift from wildfire features (posterior mean)",
    x = "P(high AQI | WSEI) - P(high AQI | WSEI=0)",
    y = "Count"
  )
```



Most mass is near 0, meaning **wildfire features don't change risk much on most days**, which is expected (most days aren't smoke-impacted).

The important part is the **right tail**: when wildfire exposure is high, `delta_risk` should become meaningfully positive — that's your “smoke episode” operational signal.

Wildfire features don't trigger false alarms on normal days, but they materially increase risk on smoke-affected days.